

Individual Incentives with Taxes.

Economic Impact Analysis of Government
Measures



Gruber, Chapter 20 (3) & 21

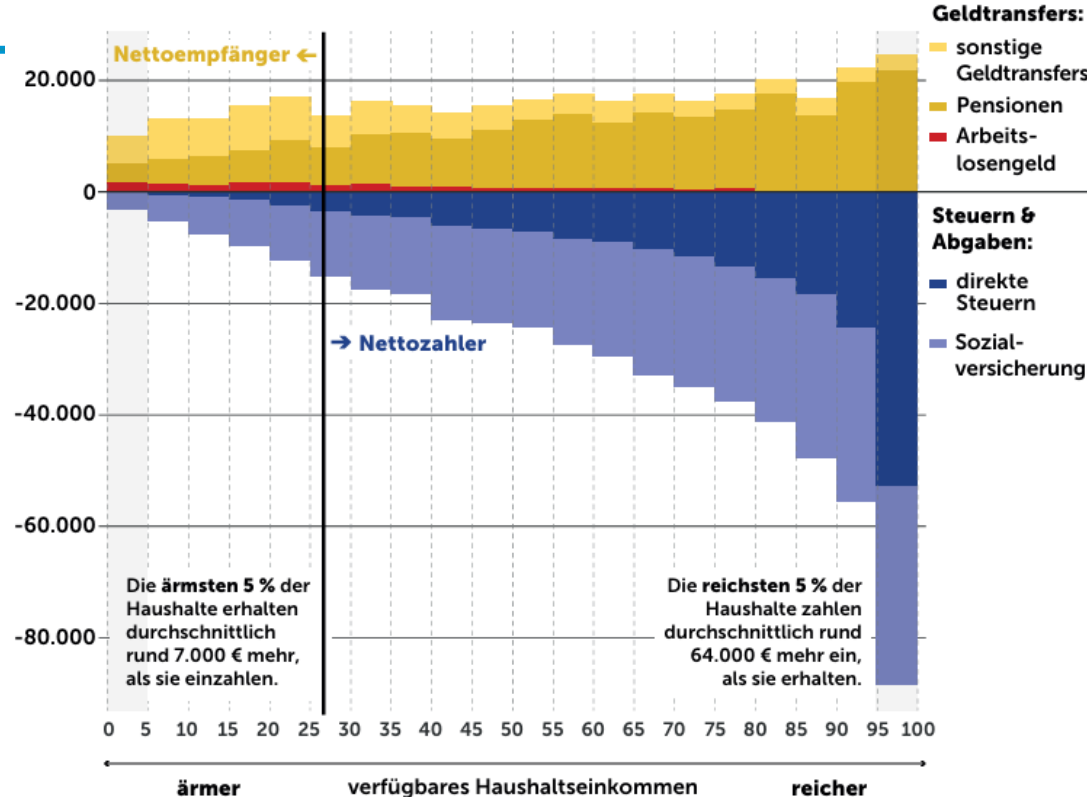


- Difference between indirect and direct taxation?
- Indirect Taxation (Goods Market)
- To levy taxes with as few distortions as possible ("targeting principle"). Direct Taxation (e.g., Income Tax)
- Second aspect: To reduce socio-economic inequalities.
- One of the most important tasks of the state!
- Optimal Progressivity is the central (and at the same time most controversial) analytical question of public finance!

Exhibit 1:

Wirkung des Sozialstaates

– direkte Geldtransfers und Abgaben nach Einkommenshöhe, in Euro



Quellen: Eigene Berechnungen, EUROMOD.

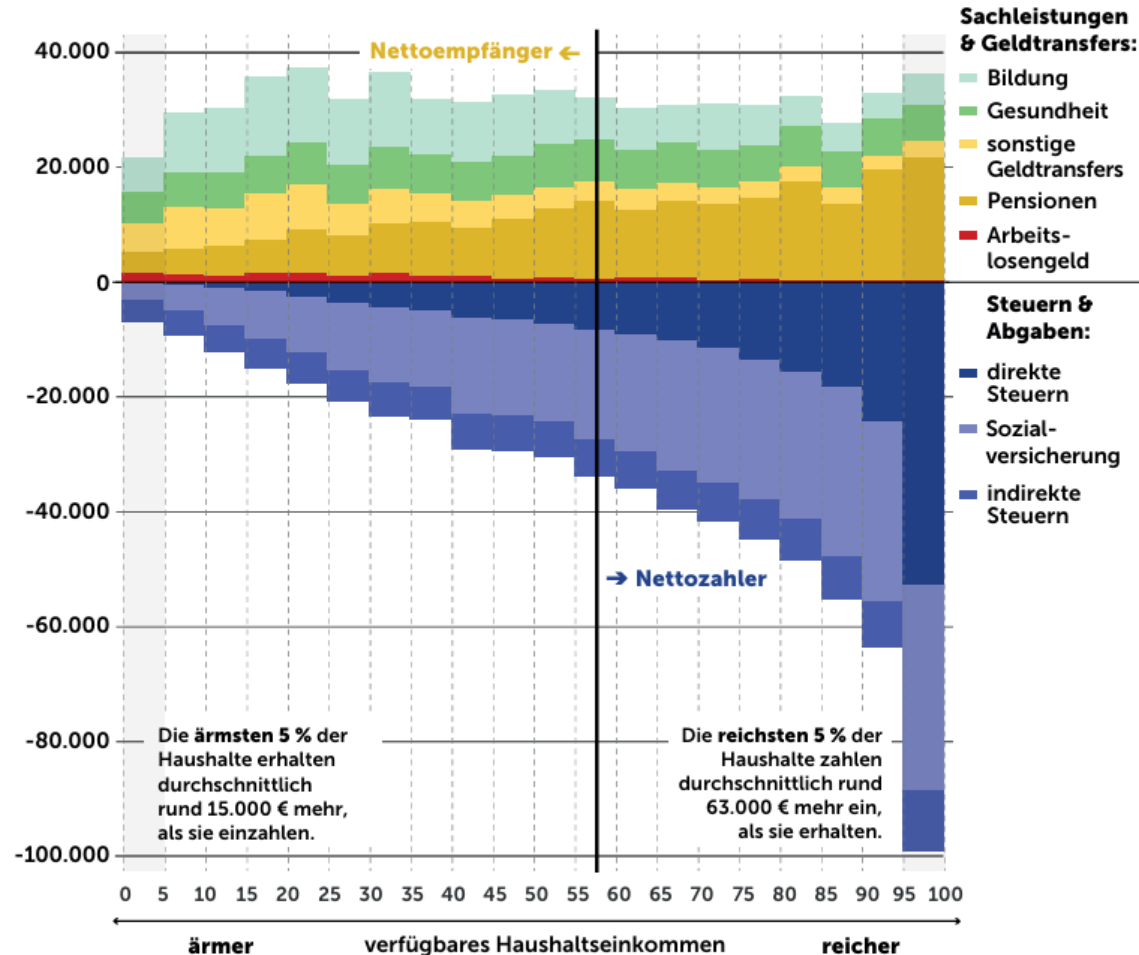
Anmerkung: Sonstige Geldtransfers umfassen Familienleistungen, Wohnbeihilfe und Geldleistungen für Gesundheit. Direkte Steuern beinhalten Steuern auf Arbeit, Kapitalerträge und

Exhibit 2:

Wirkung des Sozialstaates mit indirekten Steuern und Sachleistungen

– direkte und indirekte Transfers und Abgaben nach Einkommenshöhe, in Euro

With
**indirect
taxes** and
**indirect
transfers**



- 2021
[Umverteilung:
Das
vollständige
Bild – Agenda
Austria](#)

Redistribution, but how?

- How do we determine the optimal degree of redistribution?
- Which factors are most relevant?
- Taxes and transfers?
- Which individual reactions can be ex
- ...



Optimal Income Tax

Simplified Model **without** Behavioral Change

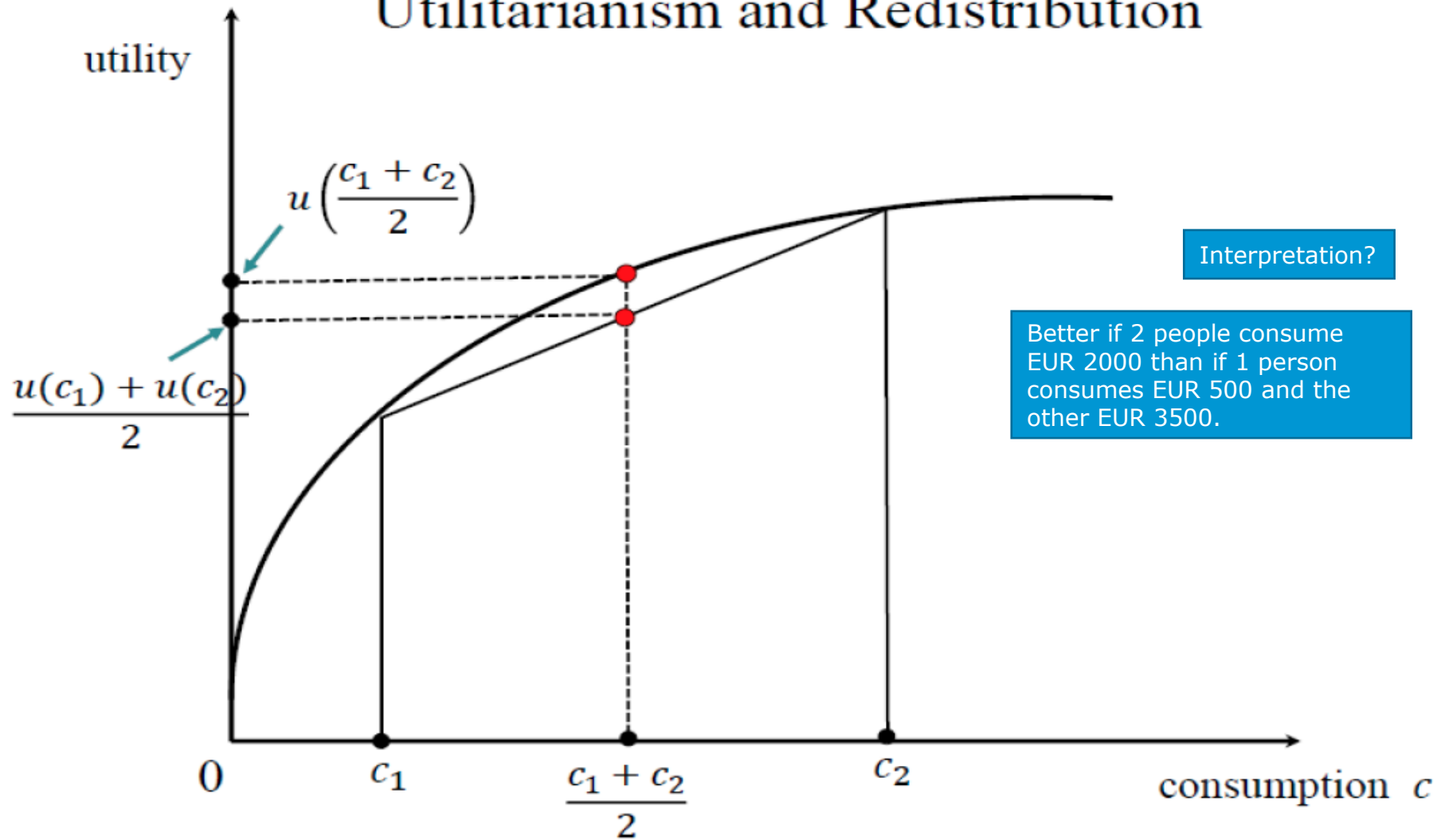
- All individuals have the same utility function
- $U_1 = U_2 = \dots = U_N$
- Utility depends on the consumption possibilities of the individuals, which are determined by **income** and the **taxes** to be paid.
- The **income** for each individual is constant. (There is no behavioral change due to the tax)
- **Utility increases** with available income; the **marginal utility decreases** with increasing income
- $\frac{\partial U}{\partial I} > 0$; $\frac{\partial^2 U}{\partial^2 I} < 0$

Optimal Income Tax

Simplified Model **without** Behavioral Change

- Society has a **utilitarian social welfare function**: $V = U_1 + U_2 + \dots + U_N$
- And maximizes this while taking into account the financing needs.
- Which tax rate maximizes the social welfare function under these assumptions?
- Marginal tax rate of 100% - everyone has the same secondary income (after taxes and transfers)
- "Principle of equimarginal sacrifice": Mill (1870), Edgeworth (1897), Ramsey (1927))
- Why?

Utilitarianism and Redistribution



What is the Marginal Tax Rate?

- How much tax you would have to pay for an additional euro of your income.

Effektiv-Tabelle 2025

Monatslohn bis *	Grenzsteuersatz
€ 1.120,00	0 %
€ 1.812,42	20 %
€ 2.997,33	30 %
€ 5.774,83	40 %
€ 8.600,33	48 %
€ 83.344,33	50 %
darüber	55 %

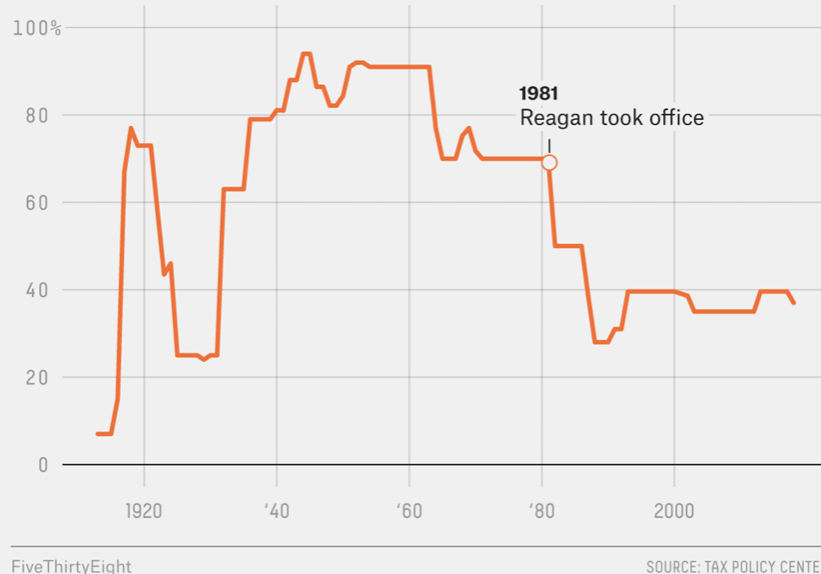
* Gehalt pro Monat ohne SV-Beitrag

- $\neq$ **Average tax rate** indicates how much tax you pay on average.
- When is the marginal tax rate = average tax rate?

100% Marginal Tax Rate Crazy?!

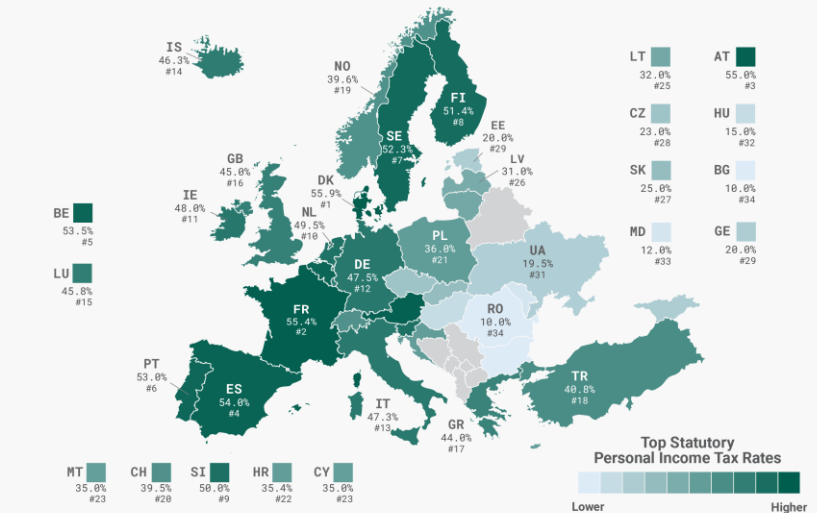
Historically, a 70 percent marginal tax rate is not unusual

The top marginal income tax rates from 1913 to 2018



Top Personal Income Tax Rates in Europe

Top Statutory Personal Income Tax Rates in 36 Major European Countries, 2024



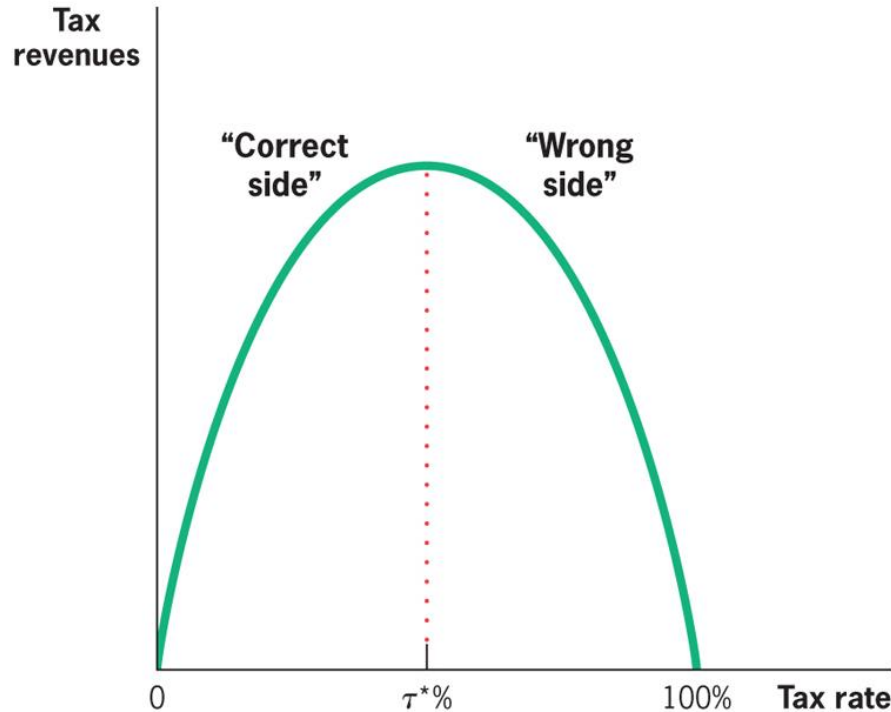
Note: Combined central and sub-central top personal income tax rates and surtaxes are shown. Social security contributions are not captured.
Source: European Commission, "Taxes in Europe Database v3," and PwC, "Worldwide Tax Summaries - Personal Income Tax (PIT) rates."

Optimal Income Tax

Simplified Model **with** Behavioral Change

- Assumption of a constant income is unrealistic!
- → Behavioral adjustment: Nobody works at 100% tax!
- → Fairness views
- Trade-off between tax fairness and efficiency (equity-efficiency trade-off)
- Second approach to optimal taxation: How big is this "trade-off"?
- If income is very elastic, then...? (Theory)
- Empirical research calculates this behavioral adjustment!

Optimal Income Tax with Behavioral Adjustment



Quelle: Gruber, 5th edition, S. 641

- With **behavioral adjustment**:
- **Higher tax $\neq$ higher revenues**
- An increase in income tax has 2 effects:
- **Mechanical**: The tax rate increases
- **Behavioral**: The tax base shrinks
- Which effect is stronger depends on the level of the tax rate!

[Hier](#) is a short assessment from our Federal President on how to interpret the Laffer curve (min 06:47-08:20)

Problems with the Laffer Curve?

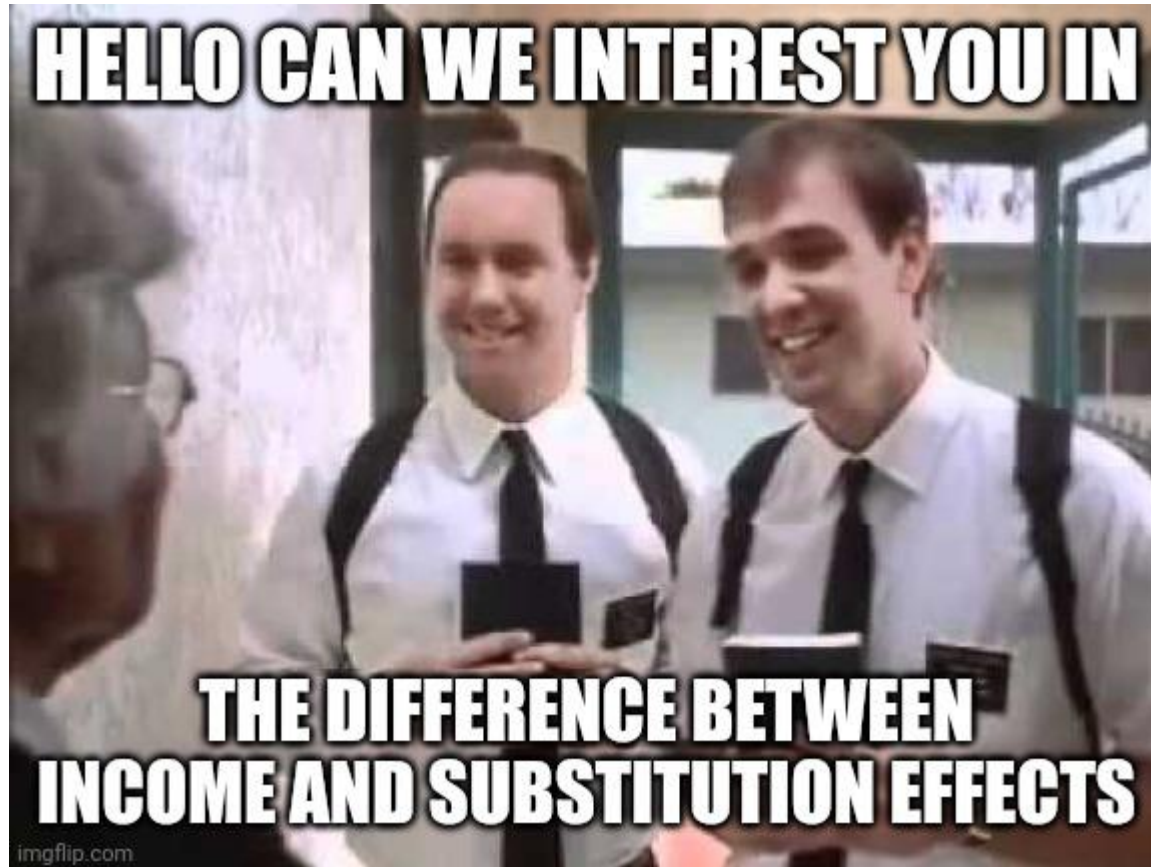
- Maximizing tax revenues vs. Efficient & Fair Tax Policy
- **Laffer Policy**: Less taxes, more spending (long-term surreal)
- **Simplified** assumptions about behavior (who reacts how?)
- No universal tax rate
- Determination of the **turning point** and **shape of the curve**.
- ...

- **Behavioral adjustments** are therefore relevant to determining the optimal progressivity (equity-efficiency trade-off).
- With which "**framework**" can we analyze these adjustments?

Incentives with Taxes and Transfers



Theorie Chapter 21



Income Tax and Labor Supply I

Theory

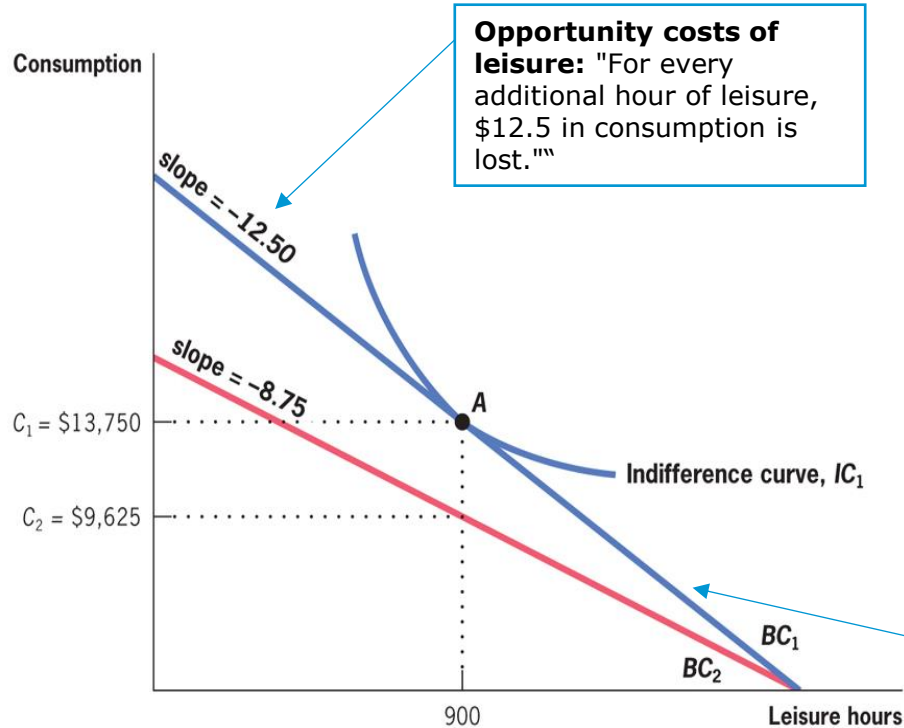
- While we have so far always considered **aggregate** decisions on markets, we will now focus on **individual** decisions.
- Individuals decide **how many of their available hours** they want to spend on work or **leisure**.



- Every hour of **leisure** reduces **labor income** (depending on the **hourly wage**) and thus consumption possibilities = budget constraint.
- Therefore, the **wage** corresponds to the **opportunity costs of leisure**.

Income Tax and Labor Supply II

Theory



- An income tax of 30% shifts the budget line inward ($BC_1 \rightarrow BC_2$).
- Hourly wage falls!
- $12.5 \times (1 - 0.3) = 8.75$
- Less can now be consumed for the same amount of work!

Budget line: All combinations above and below this can be consumed.

Quelle: Gruber, 3rd edition, S. 659

Income Tax and Labor Supply III

Theory

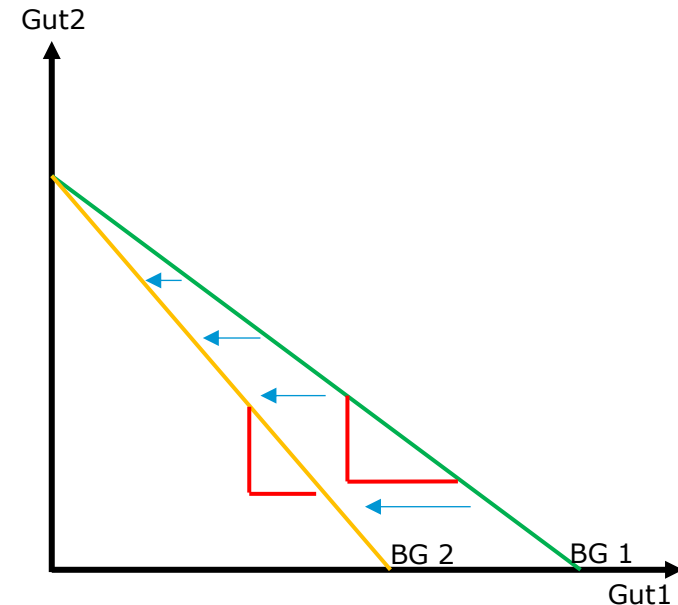
How does individual labor supply react to the tax?

→ 2 Effects

One step back

Price Changes - 2 Effects

- Starting from the goods market: A person has the choice between Good 1 and Good 2.
- → Price of Good 1 increases (e.g., due to a tax)
- → 2 effects on consumption of Good 1:
- The price ratio of Good 1 and Good 2 changes → **The budget line becomes steeper.**
 - **Substitution effect:** Good 1 becomes relatively more expensive → Consumption of Good 1 decreases relative to Good 2.
- The person can buy fewer goods overall → "Available" income/purchasing power decreases → The budget line lies further inward in (almost) all points.
 - **Income effect:** I can afford less overall → Consumption of Good 1 and 2 decreases.



- **Substitution Effect**
- Describes the change in consumption that arises due to a **change in the price ratio**.
 - = relative change in demand for Good 2 due to a price change in Good 1.
- **Income Effect**
- Describes the change in consumption that relates to the **change in purchasing power**.
- Purchasing power decreases when income remains the same and prices increase.

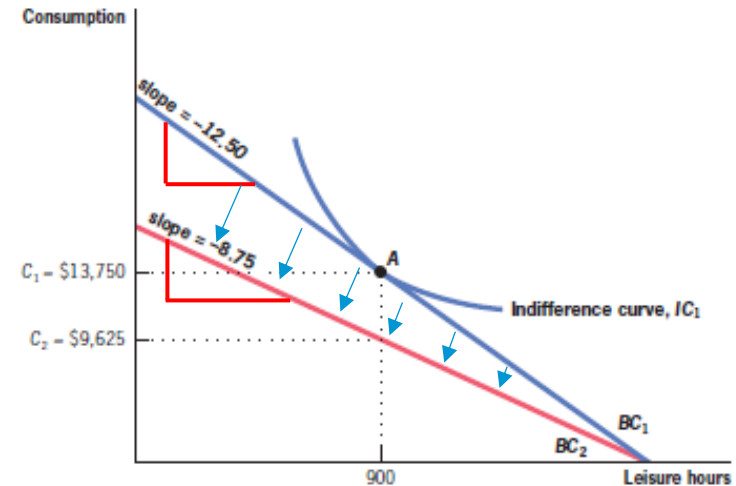
<https://inomics.com/de/terms/substitutionseffekt-und-einkommenseffekt-1428513>

Simple Example

- Ava works just to buy a coffee every day.
- She earns EUR 10 per hour, a coffee costs EUR 5, so she works 3.5 hours a week.
- If the state levies a 20% tax on her earnings, her after-tax wage falls to EUR 8 per hour.
- Now Ava has to work 4.4 hours a week to afford her coffee.
- She works harder, even though her after-tax wage has fallen, because she has a specific **income goal** that she wants to achieve.
- Which effect dominates?
- In this case, the **income** effect outweighs the **substitution** effect, and the amount of labor supplied increases.



- Effect of introducing an income tax on labor supply:
- The "**price ratio**" of consumption and leisure changes → The budget line becomes flatter.
 - **Substitution effect:** Price of leisure (opportunity costs of the foregone wage) decreases → **Consumption of leisure increases, which causes labor supply to decrease.**
- "**Available**" income decreases: For the same working hours, less net income is received → The budget line is further down.
 - **Income effect:** less "available" income
 - → can consume less of all goods (including leisure!).
 - Consumption of leisure decreases, **labor supply increases.**
 - Or: I want to maintain my income level from before the tax (standard of living) → **must work more.**

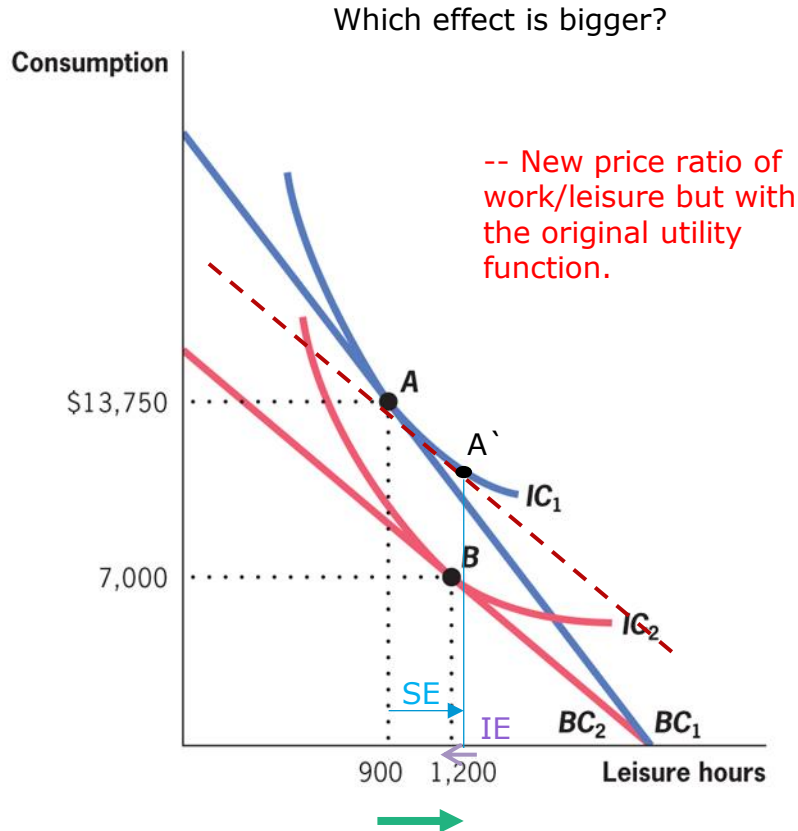


Income Tax and Labor Supply III

Theory

- How does individual labor supply react to the introduction of an income tax? → 2 Effects
- **Substitution effect** leads to an increase in leisure consumption: The opportunity costs of leisure fall because the hourly wage falls → It becomes cheaper (less wage is forgone) to consume leisure, therefore more leisure is "consumed" & less work is done.
- **Income effect** leads to more labor supply: The individual gets less income per hour of work (net hourly wage falls) and can therefore consume less of all goods (including leisure!) → more work is done.
 - Intuition for income effect: There is a certain income level that a person does not want to fall below (standard of living).
- **Which effect dominates depends on...**
 - **...the preferences (utility function) of the person!**

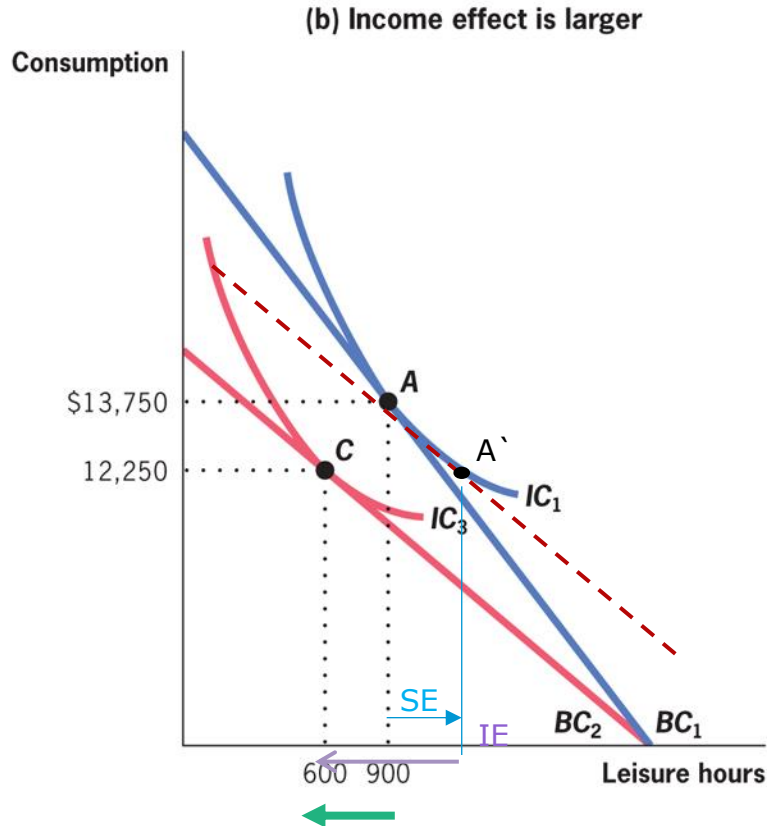
Income Tax and Labor Supply IV



- The substitution effect is larger than the income effect.
- More leisure is consumed and less work is done.
- This corresponds to the assumptions of the labor supply curves that we have considered so far → an upward-sloping labor supply curve (higher wage → more work).
- Ava works less. Coffee is not worth working for anymore.

Quelle: Gruber, 5th edition, S. 661

Income Tax and Labor Supply V



- The **income effect is larger**.
- **More work** is done and less leisure is consumed.
- Here we would have a **falling labor supply** curve → with a higher wage, less work is done!
- With a **high income** → higher loss due to proportional tax → income effect can dominate → more work (Ava works more for her coffee).
- Also with a **very low income**, the income effect can dominate (subsistence level).

Income Tax and Labor Supply VII

Example

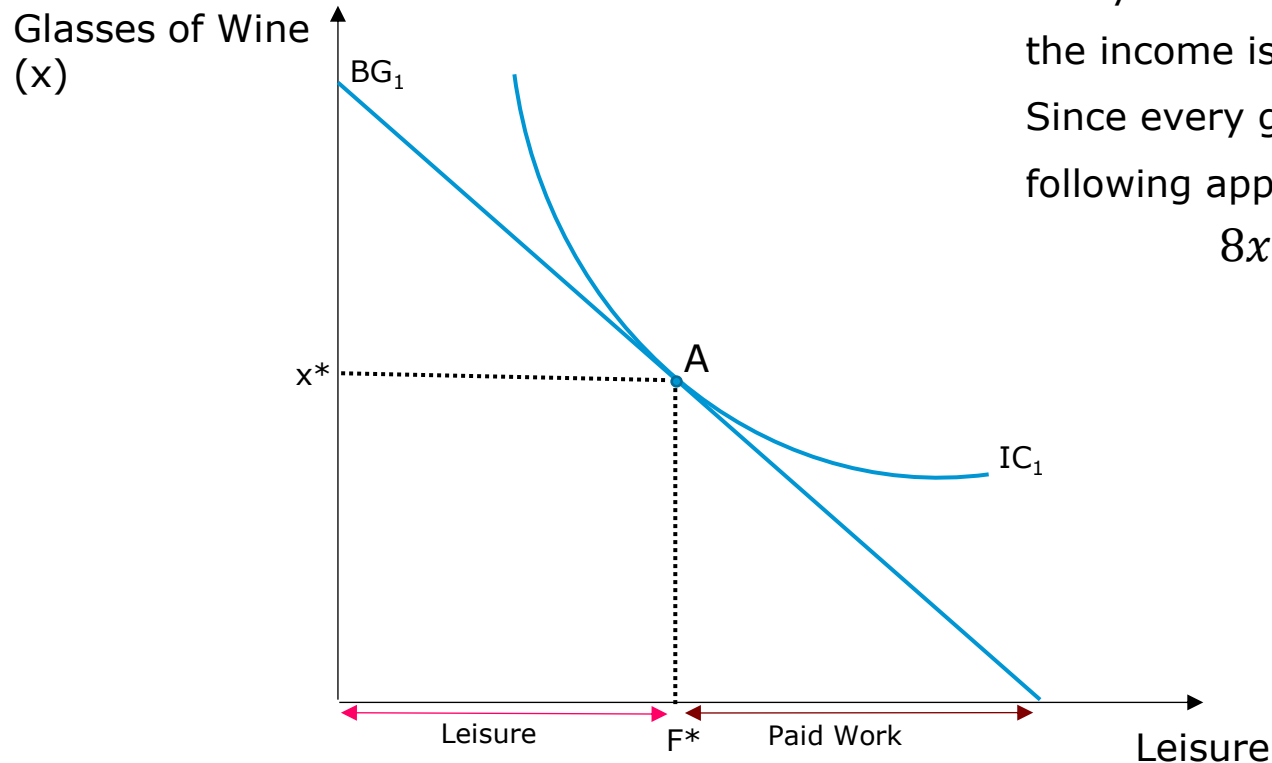


- Justus is a graduate of WU Vienna and has accepted a job at a consulting company where he earns **€12 per hour**. The job offers full flexibility: Justus can work between **0 and 60 hours per week** and thus decide for himself how to divide his time between work and leisure. His weekly income depends directly on his working hours.
- Justus spends his income exclusively on wine. A glass of wine costs **€8**, and his consumption is determined by the number of glasses (x). His preferences between **wine (in glasses)** and **leisure F (in hours)** are described by the utility function:

$$U = x^{0.6} * F^{0.4}.$$

- The state levies a **proportional income tax** with a tax rate of **20%**.
- Determine the optimal allocation of time between work and leisure as well as the utility of Justus, **without tax!**
- 1st Step Set up the budget line

Sketch + Budget Line Calculation



Every hour of work brings Justus €12, so the income is calculated by $(60-F)*12$. Since every glass of wine costs €8, the following applies:

$$8x = (60 - F) * 12$$

Income Tax and Labor Supply VII

Example without tax

Working
hours

1. Step: Set up the budget line: $8x = (60 - F) * 12$

$$x = 90 - \frac{3}{2}F$$

Relation w/p_x ($= 12/8 = 3/2$,
Wage/Price of Consumption)

→ Set up the Marginal Rate of Substitution $(MRS)_{F,x}$

Income Tax and Labor Supply VII

Example without tax

Working
hours

1. Step: Set up the budget line: $8x = (60 - F) * 12$

$$x = 90 - \frac{3}{2}F$$

Relation w/p_x ($= 12/8 = 3/2$,
Real wage
 $\Rightarrow$ For every hour of work,
1.5 glasses of wine!

2. Step: Determine the Marginal Rate of Substitution:

$$U = x^{0.6} * F^{0.4}$$

$$GRS_{F,x} = -\frac{\frac{\partial U}{\partial F}}{\frac{\partial U}{\partial x}} = -\frac{0,4 * x^{0,6} F^{-0,6}}{0,6 * x^{-0,4} F^{0,4}} = -\frac{2x}{3F}$$

We calculate the respective marginal utilities and divide them.

$\rightarrow$ Equate MRS with the slope

Income Tax and Labor Supply VII

Example without tax

Working
hours

- Step: Set up the budget line: $8x = (60 - F) * 12$

$$x = 90 - \frac{3}{2}F$$

Relation w/p_x ($= 12/8 = 3/2$,
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- Step: Determine the Marginal Rate of Substitution : $MRS_{F,x} = -\frac{\frac{\partial U}{\partial F}}{\frac{\partial U}{\partial x}} = -\frac{0,4 * x^{0,6} F^{-0,6}}{0,6 * x^{-0,4} F^{0,4}}$

$$MRS_{F,x} = -\frac{2x}{3F}$$

We calculate the respective marginal
utilities and divide them by one
another.

- Step: $MRS = \text{Slope of the budget line:}$

$$-\frac{MU_F}{MU_x} = -\frac{w}{P_x}$$

$$-\frac{2x}{3F} = -\frac{3}{2}$$

$$x = \frac{9}{4}F$$

In the household's equilibrium, the
ratio of the marginal utilities of two
goods must match the ratio of their
respective prices.

MRS: How much consumption (x)
an individual is willing to give up to
get an additional unit of leisure (F)
(hence the minus sign).

Budget line: Opportunity costs:
How much consumption one hour
of work brings in (real wage).

→ Insert X into the budget line

Income Tax and Labor Supply VII

Example without tax

Arbeitsstunden

- Step: Set up the budget line: $8x = (60 - F) * 12$

$$x = 90 - \frac{3}{2}F$$

Relation w/p_x ($= 12/8 = 3/2$,
Wage/Price of Consumption)

- Step: Determine the Marginal Rate of Substitution: $MRS_{F,x} = -\frac{\frac{\partial U}{\partial F}}{\frac{\partial U}{\partial x}} = -\frac{0,4 * x^{0,6} F^{-0,6}}{0,6 * x^{-0,4} F^{0,4}}$

$$GRS_{F,x} = -\frac{2x}{3F}$$

- Step: $MRS =$ Slope of the budget line:

$$-\frac{MU_F}{MU_x} = -\frac{P_F}{P_x} \rightarrow -\frac{2x}{3F} = -\frac{3}{2} \rightarrow x = \frac{9}{4}F$$

We calculate the respective marginal utilities and divide them by one another.

Point of tangency -> Marginal utility of leisure relative to consumption corresponds to wage (price of leisure) relative to price of consumption.

- Step: Insert x from 3. into the budget line:

$$\frac{9}{4}F = 90 - \frac{3}{2}F$$

$$\begin{aligned} F &= 24 \\ x &= 54 \\ U &= 39.04 \end{aligned}$$

Quick Recap on Household Optimum:

- MRS = Slope of the indifference curve.
- How do we get the slope = We want to stay on the curve!
- How do we do that?
 - Set the total differential of the utility function to 0!

$$\begin{aligned}dU(x_1, x_2) = 0 &\Rightarrow \frac{\partial U(x_1, x_2)}{\partial x_1} \cdot dx_1 + \frac{\partial U(x_1, x_2)}{\partial x_2} \cdot dx_2 = 0 \\&\Rightarrow \frac{dx_2}{dx_1} = -\frac{\partial U(x_1, x_2)/\partial x_1}{\partial U(x_1, x_2)/\partial x_2} = MRS_{x_1, x_2}\end{aligned}$$

Quick Recap on Household Optimum:

- **Lagrange:** Jetzt sei die Zielfunktion $U = U(x, y)$ und die Nebenbedingung $E = p_x x + p_y y$ (also ein Maximierungsproblem). Mit der vorgestellten Methode:

$$\Lambda = U(x, y) + \lambda (E - p_x x - p_y y)$$

$$\Lambda_x = U_x - \lambda p_x$$

$$\Lambda_y = U_y - \lambda p_y$$

Aus (8) und (9) folgt:

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}$$

- It also follows that MRS: $-\frac{U_x}{U_y} = -\frac{p_x}{p_y}$

Income Tax and Labor Supply VII

Example

- Justus earns **€12 per hour**. The job offers full flexibility: Justus can work between **0 and 60 hours per week** and thus decide for himself how to divide his time between work and leisure. His weekly income depends directly on his working hours.
- Justus spends his income exclusively on glasses of wine. A glass of wine costs **€8**, and his consumption is determined by the number of glasses (x). His preferences between **wine x (in glasses)** and **leisure F (in hours)** are described by the utility function:

$$U = x^{0.6} * F^{0.4}.$$

Der Staat erhebt eine **proportionale Einkommenssteuer** mit einem Steuersatz von **20%**.

Determine the optimal allocation of time as well as the utility of person A **considering the tax!**

What changes?

→ Only the budget line changes due to the tax,
→ MRS remains the same (utility function remains the same!)

Income Tax and Labor Supply VII

Example with tax

- Step: Set up the budget line with tax: $8x = (60 - F) * 12 * (1 - 0.2)$

$$x = 72 - \frac{6}{5}F$$

w is reduced by the tax

- Step: Determine the Marginal Rate of Substitution: Does not change due to the tax, as it does not affect the utility function!

$$GRS_{F,x} = -\frac{2x}{3F}$$

new Relation $w_n/p_x = (12 * 0.8) / 8$
(Price of leisure/Price of consumption)

- Step: MRS = Slope of the budget line:

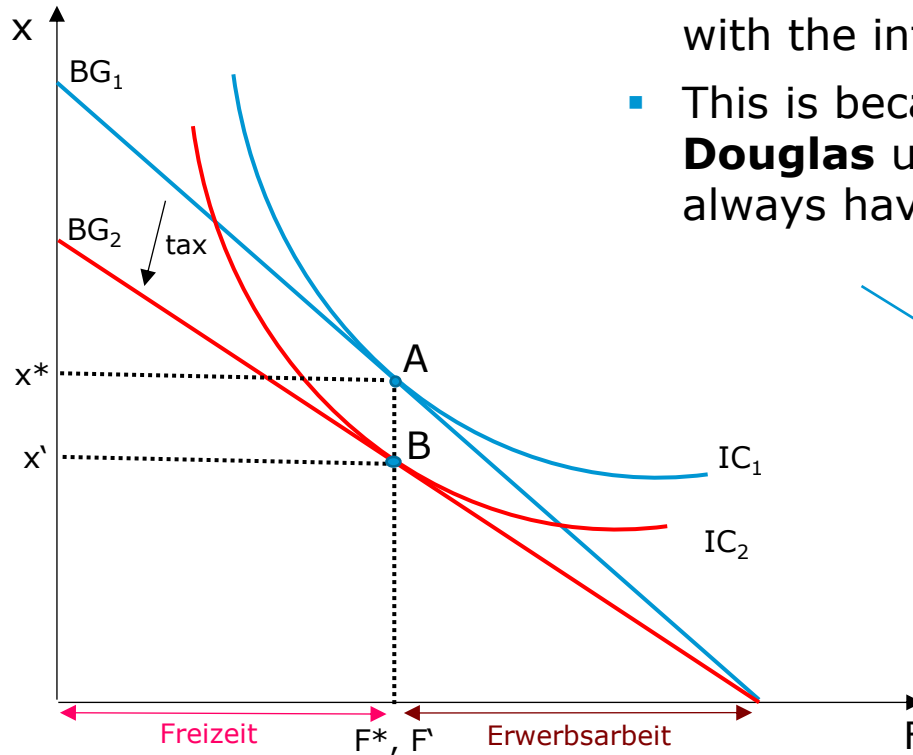
$$-\frac{2x}{3F} = -\frac{6}{5}$$

$$x = \frac{9}{5}F$$

- Step: Insert x from 3. into the budget line:

$$\frac{9}{5}F = 72 - \frac{6}{5}F$$

$$\begin{aligned} F &= 24 \\ x &= 43.2 \\ U &= 34.15 \end{aligned}$$



- Note: The working hours do not change with the introduction of the tax! Why?
- This is because we are using a **Cobb-Douglas** utility function – it doesn't always have to be this way!

$$U(x_1, x_2) = x_1^a * x_2^{1-a}$$

Constant returns of scale

Empirical Evidence



Wie reagiert das Arbeitsangebot?

Income Tax and Labor Supply

Empirical Evidence

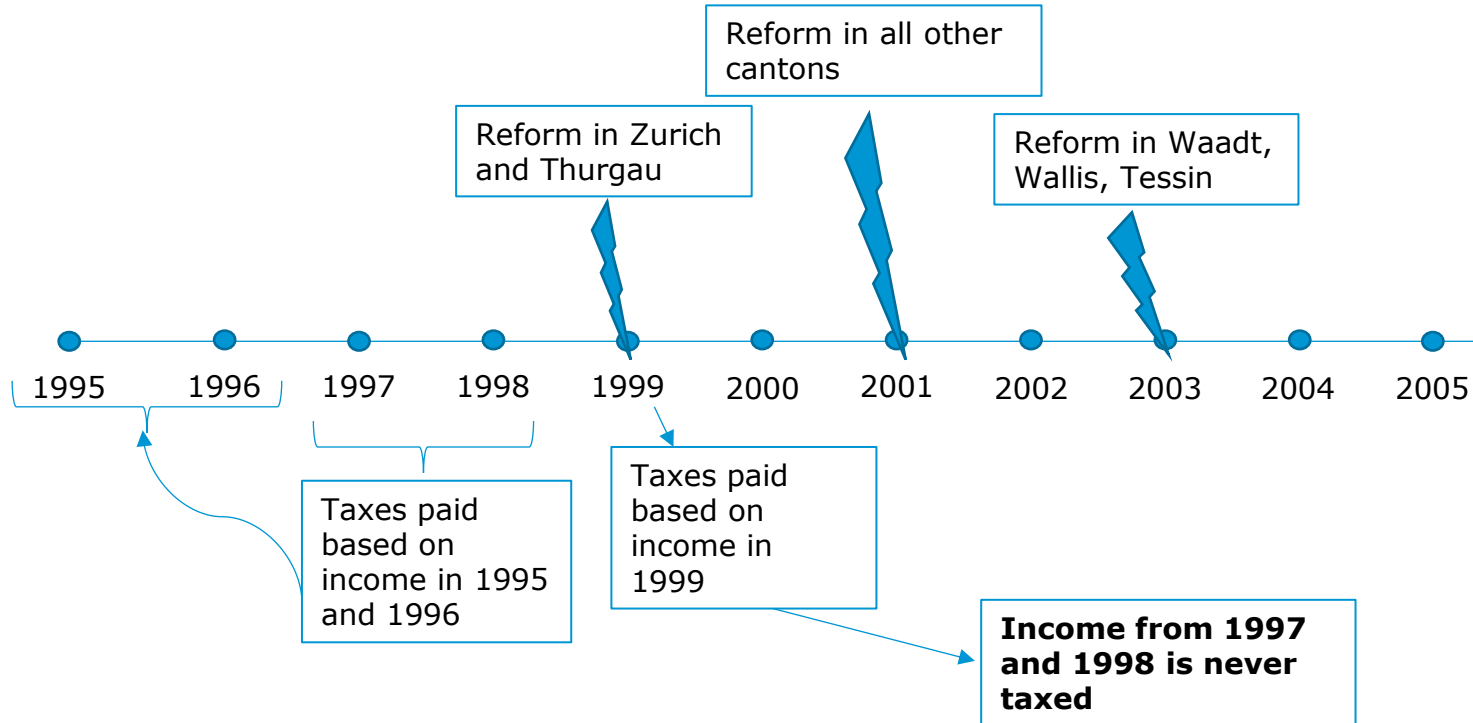
- As we have seen, the **optimal tax rate** depends on the **elasticity of the labor supply** (= slopes graphically)!
- How do employees react to taxes?
- How can the elasticity be estimated?
- Analysis based **on cross-sectional data**: Comparison of working hours of those with high tax rates and those with low tax rates
- **Quasi-experimental evidence** (natural experiments)
- e.g. Changes in the tax system: Switzerland

Income Tax & Labor Supply I

Martinez, Saez, Siegenthaler (2018)

- Transition of the **income tax system** in the cantons of Switzerland
- **three transition phases:** 1999, 2001, 2003
- **Quasi-natural experiment:** gradual introduction across different cantons → temporal trends can be controlled
- **Before the reform:** Income tax is calculated based on the average income of the previous 2 years
- **Reform:** Income tax is calculated based on current income → taxes are paid continuously (similar to Austria)

Reform – „Tax holiday“



Income Tax & Labor Supply II

Martinez, Saez, Siegenthaler (2021)

- How do labor supply and wages react to the "tax holiday"?
- **Prediction:**
- **Substitution effect:** Work becomes more attractive because there is no tax on this work => **Work more.**
- **Income effect:** Should not exist, because taxes must be paid every year!
- **Tax avoidance:** Shift income from other years to this year!

Income Tax & Labor Supply II

Martinez, Saez, Siegenthaler (2021)

- How do labor supply and wages react to the "tax holiday"?
- **Results**
- **either work, or not** (how much is worked?)
 - entweder arbeiten, oder nicht
- **Very small elasticities** → labor supply (extensive & intensive margin) as well as income react hardly to tax changes.
 - wieviel wird gearbeitet
 - Effect is driven by the **Top 5%** earners and **self-employed** (only men react: work more)
 - Effects are stronger for the self-employed
- Indication that these effects are also more a matter of **tax avoidance** than real labor supply adjustments.
 - Effects are increasingly visible for later "tax holidays" → **Learning effects?**

Income Tax and Labor Supply

Empirical Evidence

- How can the elasticity be estimated?
- Analysis based on cross-sectional data
- Quasi-experimental evidence (natural experiments)
 - e.g. Changes in the tax system: Iceland, Switzerland
- Bunching around kink points (**Kink points in the tax system**)
 - e.g. Bastani, Selin (2014)

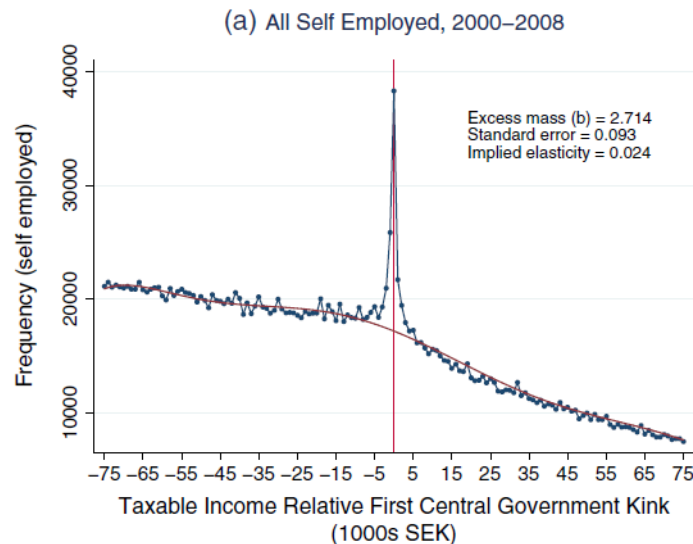
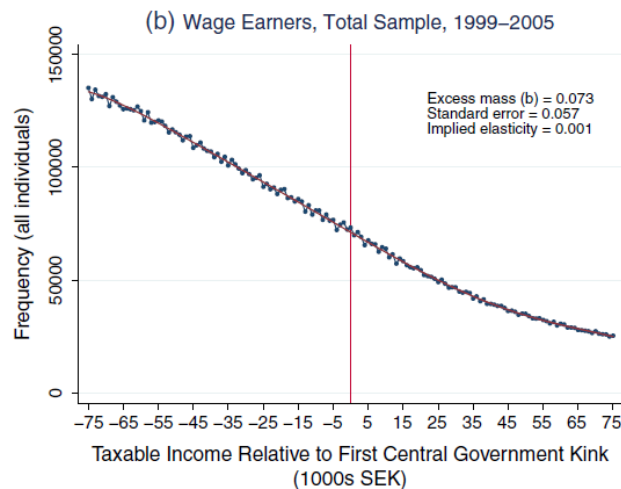


Tax brackets, for
example

Bunching around kink points

Bastani, Selin (2014)

- Search for **nonlinearities** in the **distribution** of reported income at **jumps in the tax system** in Sweden.
- Adjustments only for the **self-employed**; indications that the effect is driven by **tax avoidance** (and not real labor supply adjustments).



Income Tax and Labor Supply

Empirical Evidence

- How can the elasticity be estimated?
- Analysis based on cross-sectional data (Endogeneity problem?!)
- Quasi-experimental evidence (natural experiments)
 - e.g. Changes in the tax system: Iceland, Switzerland
- Bunching around kink points
 - e.g. Bastani, Selin (2014)
- **Experimental evidence** (randomized field experiments)
 - e.g. Basic Income Experiments

Income Tax and Labor Supply I

Household Level

- Think of a family household (primary and secondary earners):
- Who reacts most strongly to a tax?

Income Tax and Labor Supply II

Household Level

- Think of a family household (primary and secondary earners):
- Who reacts most strongly to a tax?

Mostly women

- → Empirically, **secondary earners** mostly react more strongly to incentives (increase working hours or enter the labor market for the first time)
- → The **difference in elasticities is decreasing**, because women are now also more tied to the labor market (more equal distribution of paid work) (see Blau, Kahn (2007))

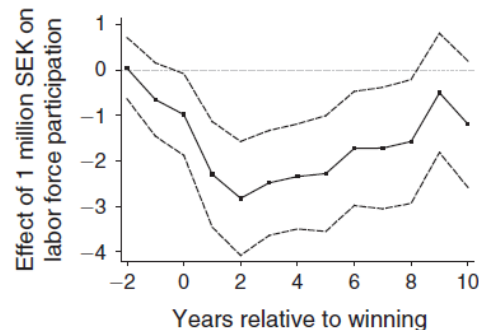
Experimental Evidence

Lottery wins – Cesarini et al. (2017)

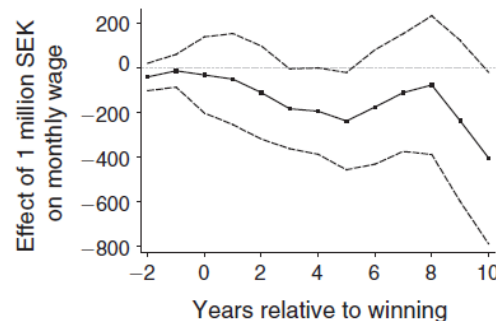
- Estimate the causal effect of a **lottery win** on labor supply at the individual and household level.
- Which effect? IE or SE?I
- → Income effect (without substitution effect)
- Find rather small, **but significant and long-term persistent negative effects** on individual labor supply.
- Extensive: Whether they completely stop working

Extensive: Ob sie komplett
aufhören zu arbeiten

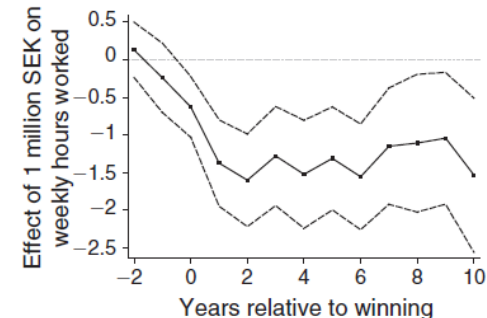
Panel A. Effect on extensive margin



Panel B. Effect on wages



Panel C. Effect on hours worked



Experimental Evidence

Lottery wins – Cesarini et al. (2017)

- BUT at the household level: **Who wins the lottery is relevant!**
- Winners reduce significantly more than spouses (regardless of whether the win is for the man or woman)
- Contradicts the assumption of the unitary household model!
- → Family and distributional policy implications for transfer policy

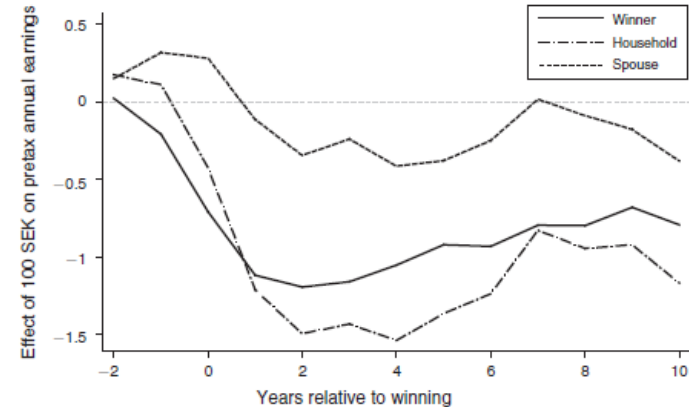


FIGURE 5. EFFECT OF WEALTH ON EARNINGS OF MARRIED WINNERS AND SPOUSES

- Paper: Investing in infants: The lasting effects of cash transfers to new families (Barr et al., 2022)
- Incentives and effects of **a special transfer payment and long-term effects on income and education.**
- What else must be considered for an **optimal transfer policy?**



- HÜ2 (Homework 2)
- Taxes and **Transfers!**